# Robotics and It's Applications

## Unit 1

1. Define a robot according to ISO and RIA. How do these definitions influence the classification of robots?
2. Explain the historical development of robotics, highlighting at least five key milestones and their significance.
3. Discuss the interdisciplinary nature of robotics. How do mechanical, electrical, and computer engineering contribute to robotics?
4. Differentiate between point-to-point and continuous path robots with suitable examples.
5. Describe the various types of joints used in robots. Illustrate with diagrams and explain their degrees of freedom.
6. What are the main robot configurations based on coordinate systems? Compare Cartesian, cylindrical, spherical, and revolute robots.
7. Explain the concepts of resolution, accuracy, and repeatability in robotics. Why are these specifications important?
8. Describe the structure and function of a robot end-effector. Classify different types of grippers and their applications.
9. Discuss the applications of robots in manufacturing, medicine, space, and agriculture. Provide specific examples for each sector.
10. Explain the economic analysis of robot installation. How is pay-back period and rate of return on investment calculated?
11. What are the key components of a robotic system? Describe the role of each component.
12. Illustrate the concept of workspace in manipulators. Differentiate between reachable and dextrous workspace with diagrams.
13. Explain the Denavit-Hartenberg (DH) parameters. How are they used in robot kinematics?
14. Describe the process of forward and inverse kinematics for a two-link planar manipulator.
15. What are roll, pitch, and yaw angles? How are they used to represent orientation in robotics?

16. Discuss the classification of robot control systems. Compare open-loop and closed-loop controls with examples.

17. Explain the significance of sensors in robotics. Classify different types of sensors used in robots.

18. Describe various robot teaching methods. Compare online and offline programming techniques.

19. What are holonomic and non-holonomic constraints? Give examples relevant to mobile robots.

20. Discuss the role of artificial intelligence in robotics. How does it enhance robot autonomy and decision-making?

21. Explain the concept of mobility and Grubler's criterion for calculating the degrees of freedom of a manipulator.

22. Compare and contrast serial and parallel manipulators. Discuss their advantages and limitations.

23. Describe the structure and working of a vacuum gripper and a magnetic gripper. Where are they typically used?

24. Explain the importance of coordinate transformations in robot kinematics. Provide an example using homogeneous transformation matrices.

25. Discuss the different drive systems used in robots: hydraulic, pneumatic, and electric. State their advantages and limitations.

26. What are the main challenges in implementing robots in hazardous environments? Suggest solutions to overcome these challenges.

27. Explain the use of robots in space exploration. What are the unique requirements for space robots?

28. Describe the process of motion planning in robotics. Why is it critical for autonomous robots?

29. Discuss the impact of robotics on employment and society. What are the ethical considerations?

30. Explain the concept of redundancy in manipulators. How does it affect the control and flexibility of robots?

31. Draw and label the different types of joints used in robot manipulators (revolute, prismatic, cylindrical, spherical, and universal joints). Explain the degrees of freedom associated with each.

32. Sketch the workspaces of Cartesian, cylindrical, spherical, and revolute (articulated) robots. Clearly indicate and differentiate between the reachable and dextrous workspace for each configuration.

33. **Given a simple 2-link planar manipulator, draw its kinematic diagram and indicate all relevant link lengths, joint variables, and coordinate frames.**

34. **Design a schematic diagram of a robotic arm capable of drawing basic geometric shapes (such as a square or circle). Label all actuators, joints, and the end-effector.**

35. **Draw a block diagram of a robot drawing system, showing the flow from input image or coordinates to the control of motors and the resulting drawing on a surface. Briefly explain the function of each block.**